

Activity using SQLite

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Course code: DSA0503

Subject : Query Processing for Data Science

Activity using SQLite Database:

**1. Create the Department Table (Parent Table) with attributes Department_ID
INTEGER PRIMARY KEY, Department_Name TEXT**

Code :

```
import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Department (

    Department_ID INTEGER PRIMARY KEY,

    Department_Name TEXT

)

""")

departments = [

    (1, "CSE"),

    (2, "ECE"),

    (3, "EEE"),

    (4, "MECH")

]

cursor.executemany("""

INSERT OR IGNORE INTO Department

(Department_ID, Department_Name)
```

```
VALUES (?, ?)

"", departments)

conn.commit()

print("Department table created successfully.")

cursor.execute("SELECT * FROM Department")

for row in cursor.fetchall():

    print(row)

conn.close()
```

Output:

```
... Department table created successfully.
(1, 'CSE')
(2, 'ECE')
(3, 'EEE')
(4, 'MECH')
```

2. Create the Student Table (Child Table) with the following attributes:

Student_ID INTEGER PRIMARY KEY,

Name TEXT,

Age INTEGER,

Department_ID INTEGER,

CGPA REAL,

FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)

Code:

```
import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("PRAGMA foreign_keys = ON")

cursor.execute("""

CREATE TABLE IF NOT EXISTS Student (

    Student_ID INTEGER PRIMARY KEY,

    Name TEXT,

    Age INTEGER,
```

```

        Department_ID INTEGER,
        CGPA REAL,
        FOREIGN KEY (Department_ID)
            REFERENCES Department(Department_ID)
    )
    """
students = [
    (101, "Arun", 20, 1, 8.5),
    (102, "Priya", 19, 1, 9.1),
    (103, "Rahul", 21, 2, 7.8),
    (104, "Sneha", 20, 2, 8.7),
    (105, "Karthik", 22, 3, 8.2),
    (106, "Divya", 21, 1, 7.9),
    (107, "Vijay", 20, 4, 8.4)
]

cursor.executemany("""
INSERT OR IGNORE INTO Student
VALUES (?, ?, ?, ?, ?)
""", students)

conn.commit()

print("Student table created successfully.")

cursor.execute("SELECT * FROM Student")

for row in cursor.fetchall():
    print(row)

conn.close()

```

Output:

```
... Student table created successfully.
(101, 'Arun', 20, 1, 8.5)
(102, 'Priya', 19, 1, 9.1)
(103, 'Rahul', 21, 2, 7.8)
(104, 'Sneha', 20, 2, 8.7)
(105, 'Karthik', 22, 3, 8.2)
(106, 'Divya', 21, 1, 7.9)
(107, 'Vijay', 20, 4, 8.4)
```

3.Display students from CSE department.

Code:

```
import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("""
SELECT S.Student_ID, S.Name, S.Age, S.CGPA
FROM Student S
JOIN Department D
ON S.Department_ID = D.Department_ID
WHERE D.Department_Name = 'CSE'
""")

print("Students from CSE Department:")

for row in cursor.fetchall():

    print(row)

conn.close()
```

Output:

```
... Students from CSE Department:
(101, 'Arun', 20, 8.5)
(102, 'Priya', 19, 9.1)
(106, 'Divya', 21, 7.9)
```

4.Count students in each department.

Code:

```
import sqlite3
```

```

conn = sqlite3.connect("college.db")
cursor = conn.cursor()
cursor.execute("""
SELECT D.Department_Name, COUNT(S.Student_ID)
FROM Department D
LEFT JOIN Student S
ON D.Department_ID = S.Department_ID
GROUP BY D.Department_ID, D.Department_Name
""")
print("Number of students in each department:")
for row in cursor.fetchall():
    print(row)
conn.close()

```

Output:

```

... Number of students in each department
('CSE', 3)
('ECE', 2)
('EEE', 1)
('MECH', 1)

```

5. Update the foreign key. For eg., change the department of a student

Code:

```

import sqlite3
conn = sqlite3.connect("college.db")
cursor = conn.cursor()
cursor.execute("PRAGMA foreign_keys = ON")
cursor.execute("""
UPDATE Student
SET Department_ID = 1
WHERE Student_ID = 103
""")
conn.commit()

```

```

print("Student department updated successfully.")
cursor.execute("""
SELECT S.Student_ID, S.Name, D.Department_Name
FROM Student S
JOIN Department D
ON S.Department_ID = D.Department_ID
WHERE S.Student_ID = 103
""")
print(cursor.fetchone())
conn.close()

```

Output:

```

... Student department updated successfully.
(103, 'Rahul', 'CSE')

```

6. Delete a department

Code:

```

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("PRAGMA foreign_keys = ON")

cursor.execute("""
DELETE FROM Student
WHERE Department_ID = 4
""")

cursor.execute("""
DELETE FROM Department
WHERE Department_ID = 4
""")

conn.commit()

print("MECH Department deleted successfully.")

```

```
cursor.execute("SELECT * FROM Department")
for row in cursor.fetchall():
    print(row)
conn.close()
```

Output:

```
*** MECH Department deleted successfully.
(1, 'CSE')
(2, 'ECE')
(3, 'EEE')
```

7. Display the student's name, student's department and CGPA if the student's CGPA is greater than 8.

Code:

```
import sqlite3
conn = sqlite3.connect("college.db")
cursor = conn.cursor()
cursor.execute("""
SELECT S.Name, D.Department_Name, S.CGPA
FROM Student S
JOIN Department D
ON S.Department_ID = D.Department_ID
WHERE S.CGPA > 8
""")
print("Students with CGPA greater than 8:")
for row in cursor.fetchall():
    print(row)
conn.close()
```

Output:

```
Students with CGPA greater than 8:
('Arun', 'CSE', 8.5)
('Priya', 'CSE', 9.1)
('Sneha', 'ECE', 8.7)
('Karthik', 'EEE', 8.2)
```